SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA1603	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3, BL4)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	20% of CIA	**Total Marks:**	100 (2 Question × 100 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	Nissanki Sri Venkata Lakshmi Kavya	**Reg. No.:**	192373036
Section / Batch:	CSE(DATA SCIENCE)	**Date:**	10-06-2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
	GRAND TOTAL:	**100 Marks**	

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

🎯 Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Decision Support System (DSS)

A **Decision Support System, or DSS**, is a computer-based system that helps people make better decisions by using data and information. It collects data, analyzes it, and presents useful information to the user. DSS is mainly used when a person needs to study different options before

making a decision. It does not make the final decision by itself; instead, it helps the user understand the situation and choose an appropriate option.

A DSS mainly has **three components: Data, Model, and User Interface**. The **Data component** stores the required information, such as sales, customer details, or business records. The **Model component** analyzes the data and helps compare different options. The **User Interface** allows the user to interact with the system and view the results in an easy way.

The working of a DSS is simple. First, the system collects the required **data** from different sources. Then, the data is analyzed using different models or methods. The results are shown to the user through the user interface, such as tables, reports, or charts. Finally, the user studies the results and makes a decision. So, the basic flow is **Data → Analysis → Results → Decision**.

For example, in a **hospital**, a DSS can help doctors or hospital managers analyze patient information, previous records, and available resources. Based on the results, they can make better decisions about treatment, hospital resources, or patient management.

The main advantage of DSS is that it helps in **faster and more informed decision-making**. It can analyze large amounts of data, compare different choices, identify useful patterns, and present information clearly. It also helps reduce the time needed to study data manually. Therefore, a Decision Support System is useful for organizations because it supports people in making effective decisions using data.

GEOTAG:



Code of Conduct

I Nissanki Sri Venkata Lakshmi Kavya(192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____